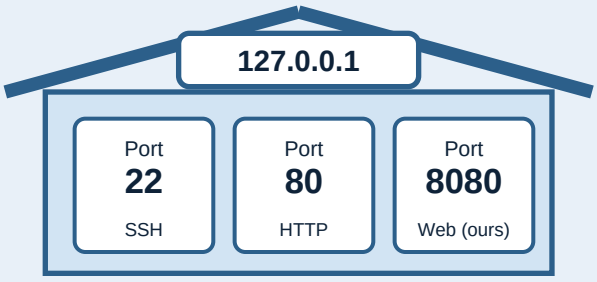
AP CYBERSECURITY

Student Cheat Guide v2: Networks, Nmap and the Terminal

Keep this open next to your lab. It explains the ideas and the commands; the lab asks you to use them.

1. The big picture: the house analogy



House = a computer

Address = its IP address

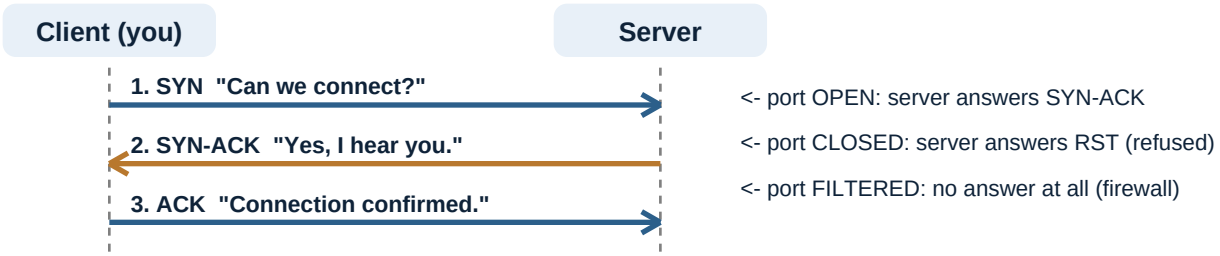
Numbered doors = ports

Someone home = a service is listening

Nmap = knocking on doors to see which open

Term	Plain meaning
IP address	The address of a computer on a network.
127.0.0.1	"This same computer" (loopback). Scanning it only touches your own VM.
Port	A numbered door (0 to 65535) that a network service uses.
Service	A program listening on a port, waiting for connections (a web server, SSH).
TCP	Connection-based: set up a connection first, then send data (web pages, SSH).
UDP	Sends data without setting up a connection (DNS lookups, streaming).
Nmap	A tool that checks which ports on a computer respond.

2. The TCP three-way handshake



Key idea: an open port means a service is listening. It does not mean the service is vulnerable. A closed port has nothing listening; a filtered port is hidden by a firewall.

3. Common ports to know

Port	Service	Port	Service	Port	Service
21	FTP (file transfer)	25	SMTP (email send)	80	HTTP (web)
22	SSH (secure remote login)	53	DNS (names)	443	HTTPS (secure web)
23	Telnet (insecure login)	8080	Alternate HTTP	3389	Remote Desktop

Page 1 = ideas to know before the lab. Page 2 = commands and what you should see during the lab. Page 3 = terminal toolbox for the log labs.

4. Nmap command card (only scan your own 127.0.0.1)

Command	What it does
`nmap -sT -p 8080 127.0.0.1`	TCP connect scan: finishes the full handshake. Works without sudo.
`nmap -sT -p 22,80,443,8080 127.0.0.1`	Scan several ports at once (commas).
`sudo nmap -sS -p 8080 127.0.0.1`	SYN scan: sends SYN, sees the reply, then does not finish the handshake. Needs sudo.
`nmap -sV -p 8080 127.0.0.1`	Try to name the service and version on that port.

Port state	What Nmap saw	What it means
open	SYN-ACK came back	A service is listening.
closed	RST came back	Reachable, but nothing is listening.
filtered	No reply	A firewall may be dropping packets.

Do not run -O, -A, decoy, or script scans on anything but your own VM, and never scan the Cyber Range network or other students. Scanning without permission can break school rules and the law.

5. Build and stop your practice server

Command	What it does
`mkdir -p ~/cybertraining; cd ~/cybertraining`	Make a folder and go into it.
`python3 -m http.server 8080 --bind 127.0.0.1`	Start a web server on port 8080 for this VM only. Keep this terminal open.
`Ctrl+C`	Stop the server (port 8080 becomes closed).
`http://127.0.0.1:8080`	Open in Firefox inside your VM to see your page.

6. What you should see

Your numbers may differ slightly. The key column is STATE.

Moment	Command	Typical result
Before the server	nmap -sT -p 22,80,443,8080 127.0.0.1	8080 closed (22 may be open or closed on your VM; 80 and 443 usually closed)
Server running	nmap -sT -p 8080 127.0.0.1	8080/tcp open
After Ctrl+C	nmap -sT -p 8080 127.0.0.1	8080/tcp closed

```
PORT     STATE  SERVICE
8080/tcp open   http-proxy
```

The service name (http-proxy, http-alt) may differ. What matters is open or closed.

7. When it goes wrong

Problem	Try this
Address already in use (server will not start)	Another server is still running on 8080. Press Ctrl+C in that terminal, or close it, then start again.
8080 shows closed while the server is running	The server terminal was closed or stopped, or you scanned the wrong address. Check it still says "Serving HTTP".
Firefox cannot open the page	Use Firefox inside the Kali VM, with http://127.0.0.1:8080 exactly. A browser on your Windows computer cannot reach it.
sudo is not available for -sS	Write down the error and watch the teacher demonstration.
nmap: command not found	Stop and tell the teacher. Do not install software without permission.
Scan seems frozen	Press Enter to see progress. Do not stop it unless told to.

8. Watch only: commands from the video

Command	What it does	Why watch only
-O	Guess the operating system	An estimate, not a guarantee; probes other machines
-A	Combines OS guess, service versions, scripts, traceroute	Noisy and slow; needs permission
-D decoy	Hides your scan among fake source addresses	Used to evade detection; do not use at school
--script vuln	Checks for known vulnerabilities (CVEs)	Only with written permission and a defined target

Rule: permission and a defined target come first.

9. Terminal toolbox (for the log labs)

Command	What it does
`ls -l | pwd | cd folder`	List files with permissions, show where you are, move into a folder.
`cat file | head -5 file`	Show a file, or only its first lines.
`wc -l file`	Count lines.
`grep "text" file`	Show lines that contain text. -c counts them, -i ignores case.
`awk '{print $NF}' file`	Print the last field of every line (for example, an IP address at the end).
`sort | uniq -c | sort -rn`	Sort lines, count repeats (needs sorted input), then rank biggest first.
`cmd1 | cmd2`	Pipe: send the output of cmd1 into cmd2.
`sha256sum -c baseline.sha256`	Check files against saved hashes. FAILED means the file changed.
`chmod 640 file`	Set permissions: owner read/write, group read, others none.

Example pipeline: `grep "Failed password" auth.log | awk '{print $NF}' | sort | uniq -c | sort -rn` counts failed logins per source address.

10. Reading permissions: -rw-r-----

Part	Who	Allowed
rw-	Owner	read, write
r--	Group	read only
---	Others (everyone else)	nothing

Octal shortcut: r=4, w=2, x=1, so rw-=6, r--=4, ---=0 and the file above is `640`. A file shown as `-rw-rw-rw-` (666) lets everyone change it. That is dangerous.

11. Is it an attack? Quick investigation checklist

Ask yourself	Why
Who? (source address, user)	Outside address at night is a red flag.
What happened? (failed, success, change)	Many failures then a success can mean a guessed password.
When? (timestamps, patterns)	Machine-fast, evenly spaced attempts point to automation.
What can I prove?	Write confirmed facts separately from guesses.
What is the next defensive step?	Contain, fix, then monitor (MFA, least privilege, logs).